

Arya Zadebrahimi

Data Scientist

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🌐 [Online Resume](#)

Professional Summary

A highly motivated bioinformatics graduate with hands-on experience in data science, software development, and computational biology. Proficient in AI fields such as machine learning, deep learning, image processing, LLMs, and generative models. Skilled in analyzing both biological and general datasets, with a strong foundation in bioinformatics algorithms and a passion for solving complex biological problems. Seeking to contribute to a dynamic team and continue growing professionally

Experience

Iran University of Medical Sciences Tehran, Iran
Intern Aug 2022

Completed hands-on training in molecular biology techniques, including cell lysis and DNA extraction, PCR-based detection of specific DNA sequences, cell culture, and microscopic cell counting.

Nobaar Tehran, Iran
Technical Lead Jun 2018 – Dec 2018

Developed two applications for customers and company drivers, and led the design and architecture of the application APIs, collaborating with the backend development team on their implementation.

Mibarim Tehran, Iran
Android Developer Oct 2017 – Mar 2018

Developed two applications supporting ridesharing services and a separate app for comparing transportation options across ridesharing platforms.

Education

Sharif University of Technology Master of Science • GPA 18.2
Bioinformatics Tehran, Iran • Sep 2021 – Jul 2024

- Applied graph neural networks (GNNs) and network analysis to predict metabolic flux and design synthetic microbial communities.
- Built and evaluated graph, text, and image models using Python and Hugging Face.
- Analyzed biological datasets with sequence analysis, RNA differential expression, and genome-scale metabolic modeling (HISAT2, SAMtools, BCFtools).
- Completed practical projects in machine learning, NLP, image processing, and generative models, including model training, evaluation, and data analysis workflows.

Islamic Azad University Bachelor of Science • GPA 15.58
Software Engineering Tehran, Iran • Sep 2015 – May 2021

- Established a solid foundation in programming, databases, and statistics through software development and computational problem-solving coursework
- Acquired proficiency in multiple programming languages, rapidly mastering new technologies as needed (e.g., learned Julia within one week to deliver a constrained optimization solution)
- Engineered an open-source Android application using Bluetooth Low Energy (BLE) to enable message sending, receiving, and broadcasting

Skills

🧬 Bioinformatics Analysis

Statistical Data Analysis, Image Processing, Mathematical Modeling, Constrained Optimization, Sequence Alignment, Differential Gene Expression Analysis, Pathway and Gene Ontology (GO) Analysis, Motif Finding / Motif Discovery Algorithms, Phylogenetic Tree Construction, scRNA Expression Analysis, FASTA/FASTQ File Processing, Molecular docking

</> AI, Coding

Python, Pytorch, R / Limma • Bioconductor, LLMs, Image Processing, Julia, ReactJS, NodeJS, Android, Java, TCL

🗄️ Databases

NCBI, UniProt, PDB, ChEMBL, PubChem, BiGG, MongoDB, MySQL, SQLite

📦 Bioinformatics Tools & Softwares

HISAT2, Samtools, BFCtools, Beagle, SHAPEIT, IGV (Integrative Genomics Viewer), Guassian, AutoDock Vina, CellProfiler, VMD

🧪 Laboratory Techniques

DNA extraction, PCR, Cell culture